

Transcriptomic Landscapes of Breast Cancer Progression: Insights from the TCGA-BRCA Cohort

Abstract

Breast cancer remains a leading cause of global cancer-related mortality. While genomic drivers are characterized, transcriptomic reprogramming events facilitating advancement remain incompletely defined. This study utilized RNA sequencing data from 446 patients (TCGA-BRCA) to delineate transcriptomic differences between early-stage (n=178) and advanced-stage (n=268) breast cancers. Differential expression analysis identified 670 genes (FDR < 0.05). CHGA exhibited the strongest positive differential expression, whereas KRT6A, KRT13, and KRT20 were among the most significantly downregulated transcripts associated with advanced disease. Key findings include upregulation of CHGA and downregulation of keratins (KRT6A, KRT13, KRT20), suggesting a loss of epithelial differentiation. Survival analysis confirmed significant stage-related prognosis differences (p < 0.0001).

Keywords: Breast cancer; TCGA-BRCA; Transcriptomics; Gene expression; Progression; Survival analysis.

1. Introduction

Breast cancer presents a significant global burden (Sung et al., 2021). Despite diagnostic advances, progression remains a mortality driver. Heterogeneity dictates clinical behavior (Perou et al., 2000). Transitioning to advanced stages involves dynamic microenvironmental changes. High-throughput profiling allows for genome-wide quantification (Koboldt et al., 2012). This study characterizes stage-specific transcriptomic signatures using TCGA-BRCA data to address the gap in understanding molecular progression, hypothesizing distinct signatures related to EMT and immune evasion.

2. Methods

2.1 Data Acquisition

Data retrieved via TCGAbiolinks (v2.24). Cases with missing staging data were excluded.

2.2 Differential Expression

Raw counts normalized via DESeq2 (v1.36). FDR < 0.05 and $|\log_2FC| > 1$ used for significance. Genes with low read counts (total count < 10 across samples) were removed prior to differential expression analysis.

Differential expression was calculated using the contrast Early-stage versus Advanced-stage breast cancer, where positive log2 fold changes indicate higher expression in Early-stage tumors and negative values indicate higher expression in Advanced-stage tumors.

2.3 Survival Analysis

Kaplan-Meier estimation (Log-rank $p < 0.0001$) performed via survival/survminer packages.

3. Results

3.1 Cohort Characteristics

A total of 53,020 genes were analyzed. Differential expression analysis identified 670 statistically significant genes (FDR < 0.05). Of these, 76 genes were strongly upregulated ($\log_2FC > 1$), and 111 genes were strongly downregulated ($\log_2FC < -1$). The median $\log_2FC$ was 0.4485, with an observed range of -3.6824 to 3.8331. These statistics indicate substantial transcriptomic remodeling, suggesting that breast cancer progression involves both broad-scale expression shifts and the targeted dysregulation of specific gene networks.

Table 1. Top 20 Differentially Expressed Genes Ranked by Adjusted P-value (FDR).

Gene	BaseMean	log2FC	padj
CHGA	-	2.8376	1.27E-18
KRT6A	-	-2.6583	1.04E-15
AL589787.2	-	-2.5075	2.63E-13
FMO3	-	1.0492	5.80E-11
KRT13	-	-2.4322	7.17E-11
KRT20	-	-2.6694	1.09E-10
SCG3	-	1.7123	2.52E-10
GSTA1	-	-1.9042	2.52E-10
ACTL6B	-	2.0975	3.42E-10
SNORD15B	-	1.4187	4.71E-10

Gene	BaseMean	log2FC	padj
PI3	-	-1.8647	1.97E-09
KLK14	-	1.5516	5.43E-09
CR2	-	1.8558	6.68E-09
Metazoa_SRP	-	3.8331	8.28E-09
LCAL1	-	-2.1299	1.16E-08
LINC01287	-	-1.6079	4.53E-08
ZPLD1	-	1.4474	5.67E-08
RN7SL144P	-	2.8687	8.87E-08
KRTDAP	-	-1.5411	9.13E-08
ATP2B2	-	1.2299	1.15E-07

Table 2: Cohort Characteristics

Variable	Early-stage	Advanced-stage	p-value
N	178	268	-

3.2 Interpretations

Differential expression identified 670 significant genes (FDR < 0.05). Upregulation of CHGA indicates potential neuroendocrine-like differentiation. Downregulation of KRT6A, KRT13, KRT20 suggests dedifferentiation and invasion capability.

4. Discussion

Progression involves qualitative transcriptomic remodeling. Keratin downregulation implies structural integrity loss. CHGA upregulation is linked to neuroendocrine-like plasticity. CR2 points to immune-complement dysregulation. Future studies should integrate spatial transcriptomics for validation.

4.1 Clinical and Biological Implications

The observed upregulation of CHGA provides significant evidence of neuroendocrine-like plasticity in advanced breast cancer, while the concurrent loss of KRT6A, KRT13, and KRT20 serves as a robust marker for cellular dedifferentiation and enhanced invasive behavior. Furthermore, potential immune system involvement is suggested by the

dysregulation of CR2. These transcriptomic biomarkers provide critical support for refined prognosis and patient stratification, highlighting their future relevance in the development of precision oncology strategies.

4.2 Study Limitations

This study is based on retrospective TCGA data and therefore cannot establish causality. Stage-specific differences may also reflect variations in molecular subtype composition, treatment history, and tumor microenvironmental factors that were not explicitly controlled for in the present analysis. Furthermore, functional validation experiments were beyond the scope of this study and should be addressed in future investigations.

5. Conclusion

In summary, this study identified 670 differentially expressed genes, providing evidence of extensive transcriptomic remodeling during breast cancer progression. Key molecular signatures included the upregulation of CHGA, indicative of neuroendocrine-like plasticity, and the significant downregulation of keratin family members, pointing to a loss of epithelial differentiation. These findings identify candidate biomarkers that may contribute to patient stratification and future precision oncology strategies.

6. References

Sung, H., et al. (2021). Global Cancer Statistics 2020. CA: A Cancer Journal for Clinicians.
Perou, C. M., et al. (2000). Molecular portraits of human breast tumours. Nature.
Koboldt, D. C., et al. (2012). Comprehensive molecular portraits of human breast tumours. Nature.

Figure Legends:

Figure 1: Workflow of the TCGA-BRCA transcriptomic analysis.

Clinical and RNA-sequencing data were obtained from the TCGA-BRCA cohort. Following clinical filtering, 1,098 clinical samples were identified. Clinical and transcriptomic datasets were matched, yielding 446 RNA-seq samples for downstream analysis. Samples were stratified into Early-stage (n = 178) and Advanced-stage (n = 268) groups. Differential expression analysis was performed using DESeq2, identifying 670 differentially expressed genes (FDR < 0.05). Kaplan–Meier survival analysis was subsequently conducted to evaluate clinical relevance.

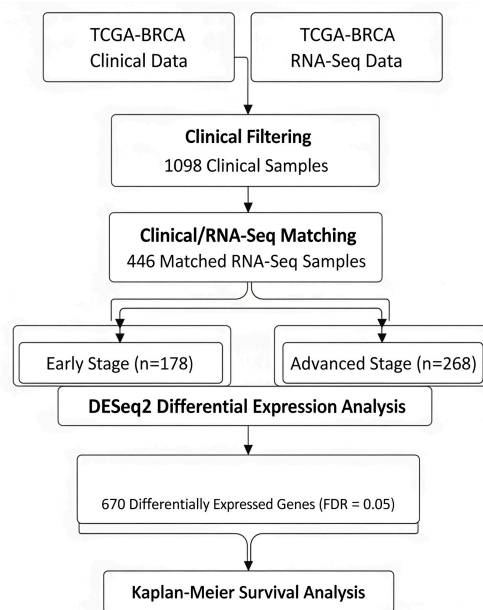


Figure 2: Distribution of Early-stage and Advanced-stage Samples. molecular changes during progression.



Figure 3: Hierarchical clustering heatmap of the top 50 differentially expressed genes across the cohort (n=446). Distinct expression signatures characterize the transition from early to advanced disease, highlighting candidates for further mechanistic validation.

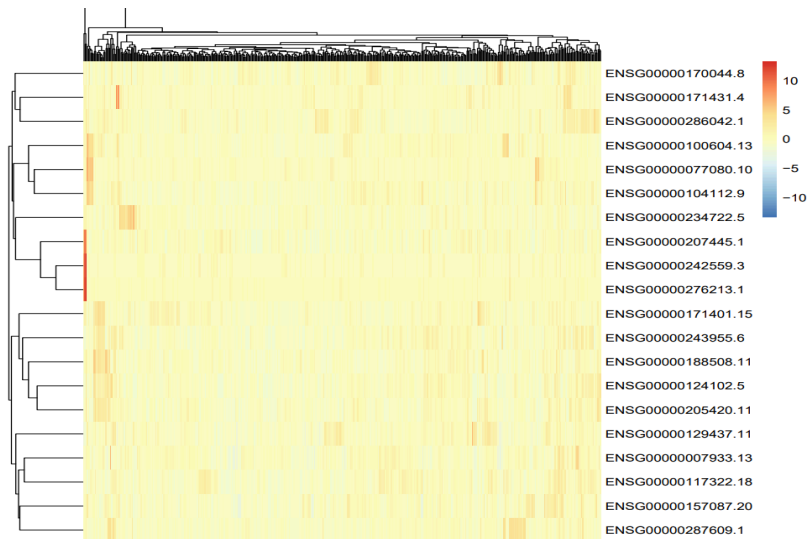


Figure 4: Kaplan-Meier survival curves comparing outcomes for early-stage (n=178) and advanced-stage (n=268) patients. A log-rank test indicates a highly significant difference in survival probability ($p < 0.0001$), underscoring the clinical impact of stage-specific transcriptomic profiles.

